The Hope: An Interactive Mobile Solution to Overcome the Writing, Reading and Speaking Weaknesses of Dyslexia

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***Abstract*— Dyslexia is the most common disorder in the world that have weaknesses in writing, reading and speaking. As a well spread disorder there are several stages of Dyslexia that has to recognize before giving the treatments. The current treatment is known as "Speech Therapy" which is given by a doctor or a therapist in a hospital or in a special education unit. These weaknesses are more wide spread and effects one in five people in the world. Dyslexia also runs in families. It can be recognized in kindergarten but if not it will be a huge disadvantage and will cause difficulties for further studies when the victim grows up. The people who are suffering from Dyslexia are step aside of the society and most of them are not very social and not open to the society. Therefor the treatment has to be creative and unique in order to get it in to the user. This tool named "The Hope" assembles Dyslexia therapies in a graphical and creative way to present the user to make it more accurate. Key content of this research covers selecting the Dyslexia stage, load the therapies and detect gestures and voice. The end result will be mainly based on Artificial Intelligence, Virtual reality, Image processing and Voice recognition technologies. Whoever suffers from Dyslexia or anyone who is willing to improve their writing, reading and speaking skills can use "The Hope" with a parent or a guardian and can get a clear improvement by following the therapies that the application provides daily. Application will provide an appreciation and a reward system to motivate and encourage the user to use the application daily. Following the therapies daily will improve the user skills. This tool lets users easily recognize their writing, reading and speaking weaknesses and will help to overcome them in a creative and accurate way.**

Keywords*— *Dyslexia, Speech Therapy, Image Processing, Gesture recognition, Mobile solution, Voice recognition, Reading and speaking weaknesses

I. INTRODUCTION

Dyslexia is a disorder, characterized by trouble with reading despite normal intelligence. Different people are affected to varying degrees. Problems may include difficulties in spelling words, reading quickly, writing words, "sounding out" words in the head, pronouncing words when reading aloud and understanding what one reads. Often these difficulties are first noticed at school.

Dyslexia is believed to be caused by both genetic and environmental factors. Some cases run in families. The underlying mechanisms of dyslexia are problems within the brain's language processing Dyslexia is diagnosed through a series of tests of memory, spelling, vision, and reading skills. Dyslexia is separate from reading difficulties caused by hearing or vision problems or by insufficient teaching [1].

Treatment involves adjusting teaching methods to meet the person's needs. While not curing the underlying problem, it may decrease the degree of symptoms. Treatments targeting vision are not effective. Dyslexia is the most common learning disability and occurs in all areas of the world. It affects 3–7% of the population; however, up to 20% may have some degree of symptoms [2]. While dyslexia is more often diagnosed in men, it has been suggested that it affects men and women equally. Some believe that dyslexia should be best considered as a different way of learning, with both benefits and downsides [3].

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Table 1: Product comparison with existing solutions

	Eye Game	Dyslexia Reading Assistant	Dyslexia Notepad	Voice Reading	The Hope
Help to read	√	√	√	√	√
Help to write	×	×	×	×	√
Help to speak	×	×	√	√	√
Attractive App	√	×	√	√	√
Letter Recognition	√	√	√	√	√
Word Recognition	√	√	√	√	√
Voice Recognition	×	×	×	√	√
Gesture Recognition	×	×	×	×	√
Improvement Showing	×	×	×	×	√
Virtual Reality	×	×	×	×	√
Image Processing	×	√	×	√	√
Camera Capture	×	√	×	×	√

II. BACKGROUND

Dyslexia is a specific reading disability in word recognition, spelling and decoding. Students with Dyslexia require special adjustments to the text and text environment to be able to read better. Current studies have shown that many assistive technologies such as software application and other tools are available in aiding Dyslexia's student in reading. However, websites that support Dyslexia's student in reading are very minimal. This discourages Dyslexia's student to use the Internet. Therefore, there is a need to identify user-interface requirements for web readability for students with Dyslexia. The requirements are font style, font size, font color, background color and spacing [4].

Dyslexia includes a large variety of literacy-related difficulties, which demands, in most cases, a personalized intervention. However, as dyslexia affects a large fraction of the population, schools cannot provide individual care for each student. The Hope application provides a tool for students to work on their literacy skills following a personalized teaching program. The design of the game and adaptation mechanisms integrated with it is aimed at maintaining student engagement for the duration of an open-ended number of playing sessions, while using a limited quantity of assets and literacy content.

By focusing on maintenance of engagement, we hope to improve learning outcomes and motivate students to also play the game outside of school [5].

People with developmental dyslexia show different levels of difficulties in psychoacoustic and psycholinguistic processing. It has been proposed that dyslexia are linked to temporal processing deficits [6]. A recent study by Ziegler et al tested the perception of global versus local pitch patterns in children with dyslexia [7]. In this study, dyslexics showed deficits in local pitch perception but not in global pitch perception, compared to the non-dyslexic controls. Such temporal processing in dyslexics has been further argued to be related to the phonological processing deficit [8]. One example is a recent study by Cheung et al who investigated the perception of lexical tones and aspiration contrasts in Cantonese in dyslexic children and he proposed that dyslexic children have lower degree of categorical perception of tones and aspiration contrasts than age-matched controls [9].

III. METHODOLOGY

The ultimate goal of executing this research project is to build up a virtual reality application for the people who are suffering from Dyslexia spread in all over the world and the main objective of the application is to provide a reliable and well accurate product to overcome the weakness of Dyslexia by using a user friendly mobile application with virtual reality.

Primarily the application will detect whether the user is in the ideal mental and intelligence level according to the age. Dyslexia has many stages. Then the application has to recognize the stage of Dyslexia by giving the user few activities. The main three weaknesses of Dyslexia are reading, speaking and writing. For detect the writing weaknesses we develop a hand glove and it connected to the application via Bluetooth. User's hand gestures are recognized by the hand glove and for the reading and speaking weaknesses, user's voice will recognize by voice recognition.

Application will run in stage by stage to overcome the weaknesses since the game load the user the stages dynamically. This whole application will run according to the "Speech Therapy" that currently use to treat Dyslexia patients completely manually. The device will guaranty that the output information is accurate and consistence. The beneficiaries of the proposed device will be the people who suffering from Dyslexia and anyone who willing to improve their reading, writing and speaking skills. Using "The Hope" application, they can overcome their weaknesses by using the application.

The application run stage by stage, when the user completes the tasks successfully, then only user can go to the next activity. Completing an activity means the user is overcome that weakness. There are mini tasks in every activity. As an example, some patients can't write 2 letters words, 3 letters words, 4 letters words etc.

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Therefore, for patients who are straggled to write 2 letters words there are different kind of activities and for 3 letters words there are different kind of activities for the user to follow. The activities and stages will arrange dynamically afterrecognizing the stage of Dyslexia of the user.

A. Detecting the stage of dyslexia

Dyslexia is a disorder and it has many stages. As an example, one who is suffering from Dyslexia may have weak in writing and fluent in speaking. Another one may have weak in reading but good in writing and speaking. Writing also can divide into many stages that are, 3 letter word weakness,4 letters word weakness, 5 letters word weakness, weak in writing sentences etc. So, detecting the stage of Dyslexia is very important to deliver the best therapy to the user.

When the user starts the application, it recognizes whether the user is in the ideal mental and intelligence stage according to the age. Application give some activities to decide the stage user like, an 8-year-old child can recognize objects and scenes.

"The Hope" supports virtual reality. It helps to recognize the object and it is a creative way to deliver an activity to a kindergarten child. Children with Dyslexia are not very social and they are like step aside of the society. They are not fluently communicating with an unknown person who is doing the therapy as the current solution to this matter. Therefore, bringing a 3D virtual world to the unsocial child is an effective way to overcome the weaknesses and get an improvement better than the current manual solution called "Speech Therapy" in hospitals, schools, special educational units etc. And treating without figuring out what stage is the user in Dyslexia, is not accurate.

First task is to decide the stage of Dyslexia. The first-time user will face an aptitude test to detect that the user can write, read and speak 3 letter, 4 letter, 5 letter and 6 letter words. After considering the performance of the user it will be compared with a well-trained predefined data set that has a 100% accuracy by using the SVM algorithm[10][11].

Figure 2 shows how the SVM algorithm that we use to train the data set to detect the Dyslexia stage works.

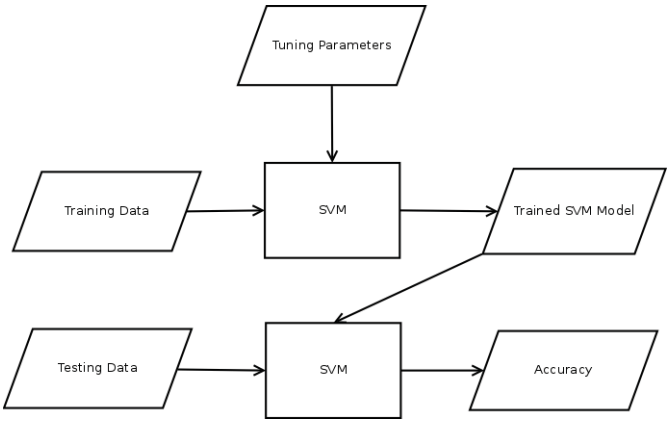


Figure2: Functionality of SVM algorithm [12]

B. Dyanamic therapy generator

After detecting the stage of dyslexia, now the user is ready for the treatment. This treatment currently doing by manually in hospitals and special educational units in front of a doctor or a therapist. Using "The Hope" application user can do it with a guardian or the parent by following the guidance of the application. After detecting the stage of dyslexia, it will go to the next step as an input.

After arriving the stage of Dyslexia, system will show the therapies that user must follow. These therapies are lining up dynamically by considering the performance of the aptitude test that the user has faced. As an example, if a user poor in writing the therapy generator give the priority to the writing therapies. If the user poor at reading therapy generator give the priority to the reading therapies.

Therapy generator make sure that the user follows the ideal and the most accurate therapy list by delivering and lining up the therapies dynamically. Therapies have sub therapies cause the dyslexia cover a wide range of disorders. By lineup and generate the therapies dynamically therapy generator makes sure that the user follows the most suitable treatment that the user wants to overcome the weaknesses of Dyslexia. User just should click the next button and follow the therapies will show the user friendliness of the application. Fig.3 shows how the therapy generator works. It shows how the Dyslexia stage become an input to the therapy generator.

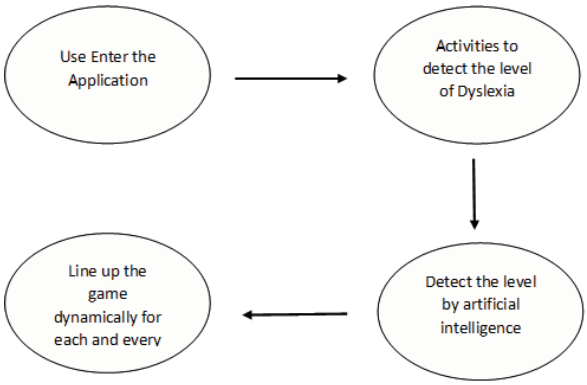


Figure 3: Work Hierarchy of therapy generator

C. Writing skills improver

Handwriting is a unique skill that a person possesses. Handwriting skill is heavily affected from dyslexia. Areas that affected by dyslexia are

- Master the rules of spelling & grammar
- Write letters numbers and symbols in the correct order
- Listen & take notes

Most of the dyslexia patients flips letters, and also patients write letters incorrect or incompletely [12].

Figure 4 shows the letters that make dyslexia patients confuse are simple letters like b, d, f, g, h, j, n, p, q, r, t, y.

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Figure 4: Dyslexic Alphabet [13]

To prevent this problems and confusion "The Hope" writing skill improver area will help out. Writing skill improver will first get the handwriting as an input. The written input will be convert to an image, image will be recognized and convert to symbolic form. This will be then compared with the predefined data set, and show the errors.

Handwriting recognition means the ability of the computer to analyze hand written document (paper, photographs, touch screen and other devices) inputs and recognize the characters, words, sentences.

Figure.5 shows the procedure of writing skills improver that use to detect the hand gestures of the user.

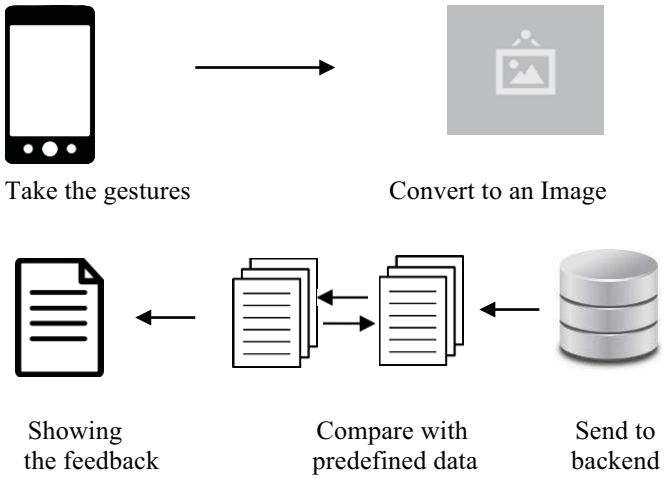


Figure 5: Procedure of writing skills improver

Hand writing is associated with recognition, interpretation. The language represented by the graphical form will be converted into symbolic form in recognition part. In Handwriting interpretation, what happen is determining the meaning of the recognized symbolic form. [14]

D. Speaking skills improver

The main motive of developing this app is to provide a support for children suffering from vocal disabilities due to dyslexia. Primarily, this app will aid doctors to assess how severely a child is suffering from dyslexia. Via the "HOPE" app, doctors will experience a faster, more efficient method of diagnosis compared to the older, slower ways. Furthermore, this app will enable parents of dyslexic children to help their children speak, in the comfort of their own homes.

HOPE has been developed focusing mainly on younger children, whereby the interfaces have all been designed in a child friendly, colorful and attractive theme that appeals to younger audiences.

The app works by providing paragraphs and words of increasing complexities for the child to read as they progress further. For example, a beginner to this app will get words like "Cat", "Dog", "Cake" etc., and as they progress further they will be introduced to words like "Come", "Running", "Swimming" etc., and finally they will be introduced to more complex words such as "Caterpillar", "Lawnmower" etc.[15]

The app recognizes individual words spoken by the child this process is instantaneous, and allows us to come to a clearer conclusion on their level of speech. If a word has been pronounced incorrectly, the app then repeats the word repeatedly till the child gets it right, but there is also a "SKIP" option in order to avoid frustration. The data gathered by the app is then presented in a graphical form for analysis by the doctors/parents. Furthermore, the app is also enabled with a function that allows doctors to add their own activities to the software.

IV. Results and Discussion

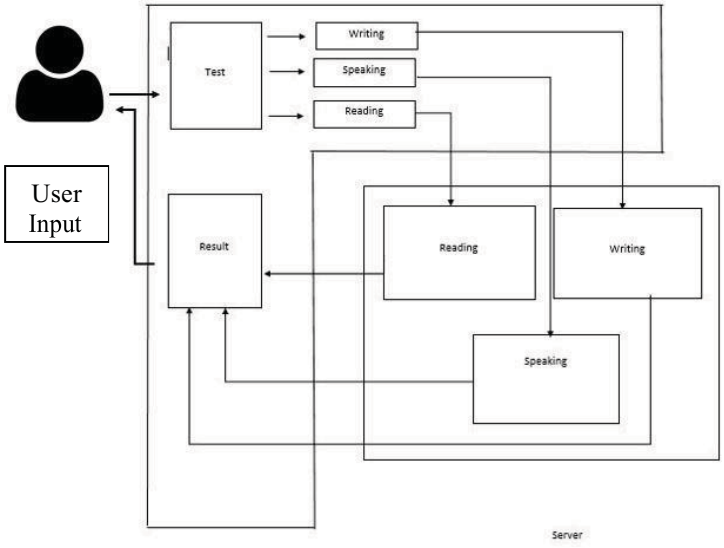


Figure 5: System Diagram

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The main result of this research is to overcome all the weaknesses of Dyslexia and improve writing, reading and speaking skills. This is an Android based mobile application. In the analysis, we used Unity game engine and MATLAB. To give a different experience to the children who using this and to be more accurate, “The hope” supports virtual reality too. Data set that we generate by using the information we collected from Lady Ridgway Hospital, was perfectly match for our requirements. We used several algorithms like SVM, SVM Linear and SVM Gaussian Etc.

In machine learning, support vector machines are supervised learning models with associated learning algorithms that analyze data used for classification and regression analysis. Given a set of training examples, each marked as belonging to one or the other of two categories, an SVM training algorithm builds a model that assigns new examples to one category or the other, making it a non-probabilistic binary linear classifier [16].

An SVM model is a representation of the examples as points in space, mapped so that the examples of the separate categories are divided by a clear gap that is as wide as possible. New examples are then mapped into that same space and predicted to belong to a category based on which side of the gap they fall. There were some problems when implementing this application [17].

Generate the data set was the first task to detect the stage of the disease. We had to visit Lady Ridgway Hospital and Victory Special education school several times. Another hard task was to detect the hand gestures and get a clear image to image processing. We developed a hand gesture glove that work as a mouse to help the user to write. Furthermore, the user can see his/her improvement in a graphical way. Application gives points to motivate the user, as we give the priority to the children who are suffering from Dyslexia. Anyone who willing to improve their reading, writing and speaking skills can use the application.

According to Artificial intelligent, image processing and voice recognition components, it was revealed that detect the stage of Dyslexia and weaknesses of Dyslexia can overcome. It provides more accurate results than the current solution, “Speech therapy”. Detecting the Dyslexia stage is 100% accurate by using SVM algorithm. User can use the hand gesture glove or use the mobile screen as a doodle. Hand gesture glove makes the application user friendly.

Speech recognition is the ability of a machine or program to identify words and phrases in spoken language and convert them to a machine-readable format. Rudimentary speech recognition software has a limited vocabulary of words and phrases, and it may only identify these if they are spoken very clearly. More sophisticated software has the ability to accept natural speech. Speech recognition works using algorithms through acoustic and language modeling. [18]

Acoustic modeling represents the relationship between linguistic units of speech and audio signals; language modeling matches sounds with word sequences to help distinguish between words that sound similar. Speech recognition performance is measured by accuracy and speed. Accuracy is measured with word error rate. WER works at the word level and identifies inaccuracies in transcription, although it cannot identify how the error occurred. Speed is measured with the real-time factor. A variety of factors can affect computer speech recognition performance, including pronunciation, accent, pitch and volume and background noise [19].

“The Hope” developed for English language and can only improve English writing, reading and speaking skills. Authors are planning to deliver a new update for Sinhala language in future.

Currently the application is running for android devices. Supporting for IOS is the next task. It will be a huge advantage for the IOS users because the application support for virtual reality too.

V. CONCLUSION

In this research paper, the researchers have presented an application to overcome the weaknesses of Dyslexia, writing, reading and speaking. The main outcome of research is to improve the writing, reading and speaking skills. Virtual Reality offers a real visualization of the final product for customers, and will be an advantage while it increases the dedication and the interest of the user. This study proves that a business plan like this is more effective and accurate than a manual process to overcome a disorder

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